

HVAC Thermostat and Zone Controller

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Overview— This project implements a dual-zone HVAC control system using an Arduino to regulate temperature independently in two zones. Each zone's temperature is read from analog sensors, compared with a user-defined setpoint, and adjusted by opening or closing servo-driven vents. LEDs indicate the system state: green for ON, red for heating, and blue for cooling. Users can turn the system on/off, switch between heating and cooling, and adjust the desired temperature through push buttons and switches. A finite state machine (FSM) controls the system's operation, managing states such as OFF, HEAT, COOL, and ERROR. Shared variables coordinate communication between temperature control, servo actuation, and LED indication. The system successfully demonstrates automated zone temperature regulation. In heating mode, vents open when the zone temperature is below the setpoint; in cooling mode, they open when it's above. Periodic serial output reports temperature, system mode, and zone status. Overall, the project achieves a reliable, responsive, and expandable foundation for smart HVAC control.

I. INTRODUCTION

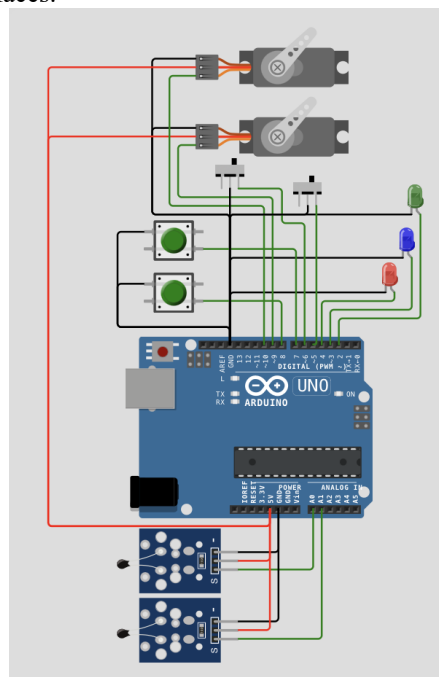
The purpose of this project is to design and implement an automated dual-zone HVAC control system that maintains user-defined temperature levels independently across two zones. This project uses an Arduino-based embedded system capable of precise, zone-specific control through real-time temperature monitoring and servo-driven air vent adjustments. The system's core is structured around a finite state machine (FSM), which simplifies the control logic by organizing system behavior into well-defined states such as OFF, HEAT, COOL, and ERROR. Each state governs specific operations—such as adjusting servo positions, activating LEDs for status indication, or handling user inputs from switches and push buttons. This modular approach enhances both the scalability and maintainability of the system. The project's objectives include creating a reliable control mechanism that efficiently regulates airflow, minimizes energy waste, and provides a clear interface for user interaction. By leveraging simple electronic components—temperature sensors, servos, LEDs, and switches—the design demonstrates how low-cost hardware can be combined with robust software architecture to achieve intelligent climate control. Ultimately, this project showcases how embedded systems can be applied to everyday automation challenges, offering an adaptable platform for future smart home or building management applications.

II. APPLICATION DESIGN

This project was developed and simulated using Wokwi, an online Arduino simulator that allowed real-time testing and debugging of the dual-zone HVAC control system. The system follows a Finite State Machine (FSM)-based design ideology combined with clear separation of logic, control, and output.

A. System Architecture

The model represents the system data, including temperature readings, setpoints, and zone states (zone1Temp, zone2Temp, setTemp, zone1Open, zone2Open). The view consists of the LED indicators and servo positions, which provide visual feedback to the user about the system's operation. The controller manages transitions between states (OFF, HEAT, COOL, ERROR), processes user inputs (switches, push buttons), and updates outputs based on the logic defined in the FSM. This separation of concerns simplifies debugging and future modifications, such as integrating additional sensors or user interfaces.

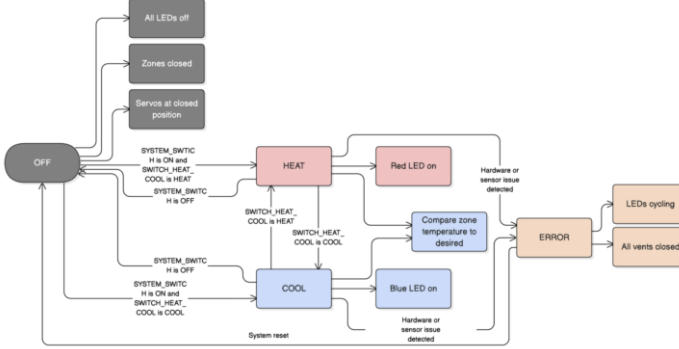


B. State-Machine

The system is modeled as two interacting state machines:

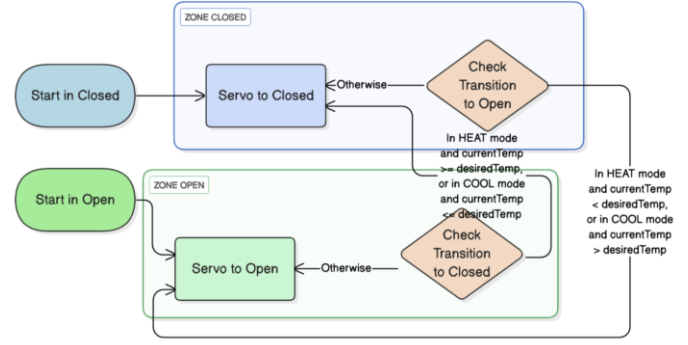
1. System Control State Machine - This state machine governs the overall operation of the HVAC system — whether it's off, heating, cooling, or in error.

The System Control State Machine is the core logic governing the overall operation of the dual-zone HVAC system. It manages the system's high-level behavior by transitioning between four main states: OFF, HEAT, COOL, and ERROR, based on user inputs and system conditions. In the OFF state, the system remains inactive with all LEDs turned off and vents closed. When the power switch is turned on, the system transitions to either HEAT or COOL mode depending on the position of the mode selector switch. In the HEAT state, the system activates the red LED and continuously compares each zone's temperature to the desired setpoint—opening vents for zones below the setpoint to allow warm airflow. Similarly, in the COOL state, the blue LED is illuminated while the system opens vents for zones exceeding the setpoint to enable cooling. Both active states can transition back to OFF when the power switch is turned off or to each other if the user changes mode. If any hardware or sensor malfunction is detected, the system transitions to the ERROR state, where all LEDs flash in sequence to alert the user, and all vents are forced closed for safety. The system remains in this state until it is manually reset by switching the system off, ensuring controlled recovery and reliability in operation.



2. Zone Control State Machine (for each zone z1 and z2)– This state machine controls vent opening/closing based on the current temperature and mode.

The Zone Control State Machine operates independently for each zone (Z1 and Z2) and is responsible for regulating airflow by controlling the vent's open or closed position based on real-time temperature readings and the system's current mode. Each zone alternates between two states — ZONE_CLOSED and ZONE_OPEN — to maintain the desired temperature efficiently. In the ZONE_CLOSED state, the servo is set to a 1 ms pulse width, keeping the vent shut to prevent unnecessary airflow. The state transitions to ZONE_OPEN when the zone requires temperature correction: in HEAT mode, if the current temperature falls below the desired setpoint, or in COOL mode, if it rises above the setpoint. Once the temperature stabilizes and reaches or crosses the desired value, the vent transitions back to the ZONE_CLOSED state, with the servo set to a 2 ms pulse width to close airflow. This cyclical control ensures that each zone maintains its target temperature without interference from the other. By using separate control logic per zone, the system achieves independent and efficient thermal regulation, reducing energy waste while maintaining user comfort across multiple areas.



III. RESULTS

The dual-zone HVAC control system performed successfully according to the project objectives. Each zone operated independently, accurately responding to temperature variations and maintaining the user-defined setpoint. When a zone's temperature dropped below the setpoint in heating mode, its corresponding vent servo opened to allow warm air circulation. Conversely, in cooling mode, the vent opened when the zone temperature rose above the setpoint. This confirmed that the control logic and finite state machine (FSM) transitions functioned correctly across all operating modes. The LED indicators clearly reflected the system's status: the green LED illuminated when the system was powered on, while red and blue LEDs indicated heating and cooling modes, respectively. The system also entered an error state correctly when triggered, flashing all LEDs simultaneously to alert the user. Serial monitoring provided continuous feedback, displaying real-time temperature readings, active mode, and vent positions. Overall, the system met all design goals—achieving accurate temperature control, reliable state transitions, and stable real-time performance. The implementation demonstrates a robust and modular solution suitable for extending into larger or more complex HVAC automation systems.

IV. FUTURE WORK

Future improvements could include adding an LCD or OLED display to show live temperatures and settings. Including wireless communication (Wi-Fi/Bluetooth) for remote control via mobile devices. Another extension could involve adding more zones or integrating humidity sensors for more comprehensive climate control.

V. DISCUSSION

The project demonstrates how embedded systems can implement real-time environmental control efficiently with minimal hardware. The FSM-based approach made debugging and logical expansion straightforward.